

PVsyst - Simulation report

Standalone system

Project: standalone without shading 24-03-2026

Variant: New simulation variant

Standalone system with batteries

System power: 410 kWp

George Town - India

**PVsyst V7.4.7**

VC0, Simulation date:
25/03/26 20:53
with V7.4.7

Project summary**Geographical Site****George Town**

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary**Standalone system****PV Field Orientation**

Fixed plane

Tilt/Azimuth 15 / 20 °

Standalone system with batteries**User's needs**

Fixed constant load

75.0 kW

Global

657 MWh/Year

System information**PV Array**

Nb. of modules

630 units

Pnom total

410 kWp

Battery pack

Technology

Lead-acid, sealed, Gel

Nb. of units

3540 units

Voltage

240 V

Capacity

35400 Ah

Results summary

Useful energy from solar 598274 kWh/year

Specific production 1461 kWh/kWp/year

Perf. Ratio PR 77.47 %

Missing Energy 58726 kWh/year

Available solar energy 626300 kWh/year

Solar Fraction SF 91.06 %

Excess (unused) 12145 kWh/year

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Main results	4
Loss diagram	5
Predef. graphs	6

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General parameters**Standalone system****PV Field Orientation****Orientation**

Fixed plane

Tilt/Azimuth 15 / 20 °

User's needs

Fixed constant load

75.0 kW

Global

657 MWh/Year

Standalone system with batteries**Sheds configuration**

No 3D scene defined

Models used

Transposition Perez

Diffuse Perez, Meteonorm

Circumsolar separate

PV Array Characteristics**PV module**

Manufacturer

AE Solar

Model

AE 650ME-132

(Original PVsyst database)

Unit Nom. Power

650 Wp

Number of PV modules

630 units

Nominal (STC)

410 kWp

Modules

35 string x 18 In series

At operating cond. (50°C)

Pmpp

375 kWp

U mpp

617 V

I mpp

607 A

Controller

Universal controller

Technology

MPPT converter

Temp coeff.

-5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies

97.0 / 95.0 %

Total PV power

Nominal (STC)

410 kWp

Total

630 modules

Module area

1957 m²

Cell area

1834 m²**Battery**

Manufacturer

Narada

Model

AcmeG 12V 200

Technology

Lead-acid, sealed, Gel

Nb. of units

177 in parallel x 20 in series

Discharging min. SOC

20.0 %

Stored energy

7005.6 kWh

Battery Pack Characteristics

Voltage

240 V

Nominal Capacity

35400 Ah (C10)

Temperature

Fixed 20 °C

Battery Management control

Threshold commands as

SOC calculation

Charging

SOC = 0.90 / 0.75

approx.

265.8 / 253.4 V

Discharging

SOC = 0.20 / 0.45

approx.

236.4 / 245.7 V

Array losses**Thermal Loss factor**

Module temperature according to irradiance

Uc (const) 20.0 W/m²KUv (wind) 0.0 W/m²K/m/s**DC wiring losses**

Global array res.

17 mΩ

Loss Fraction

1.5 % at STC

Serie Diode Loss

Voltage drop

0.7 V

Loss Fraction

0.1 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction

2.0 % at MPP

Strings Mismatch loss

Loss Fraction

0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



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Main results

System Production

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Available solar energy 626300 kWh/year
Excess (unused) 12145 kWh/year

Perf. Ratio PR 77.47 %
Solar Fraction SF 91.06 %

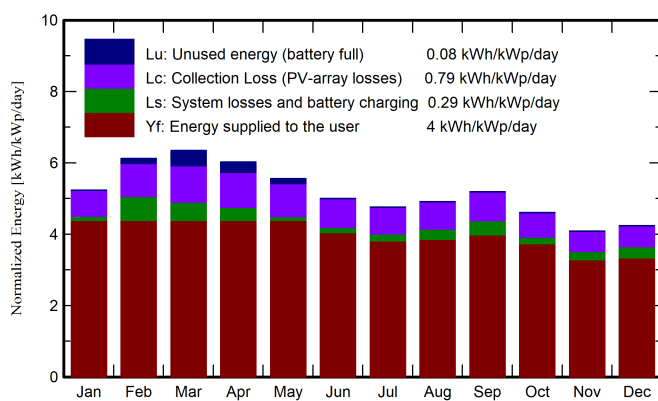
Loss of Load

Time Fraction 0.0 %
Missing Energy 58726 kWh/year

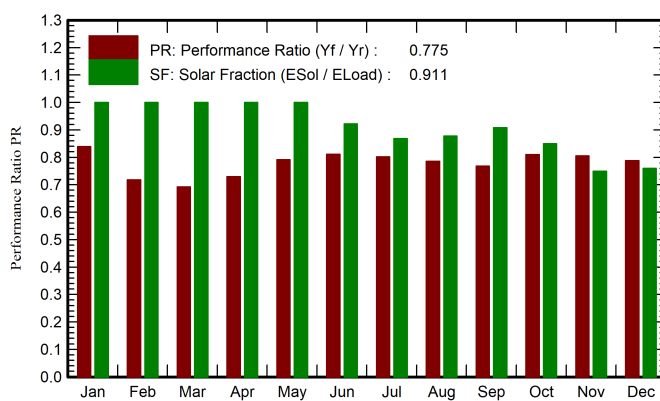
Battery aging (State of Wear)

Cycles SOW 96.5 %
Static SOW 91.7 %
Battery lifetime 12.0 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	144.2	158.9	54882	0	0	55800	55800	1.000
February	156.4	168.0	57261	1497	0	50400	50400	1.000
March	189.7	192.8	65078	5389	0	55800	55800	1.000
April	184.7	176.8	59563	3530	0	54000	54000	1.000
May	182.6	168.1	56469	1723	0	55800	55800	1.000
June	162.3	146.1	49280	2	4213	49787	54000	0.922
July	158.0	143.6	48715	0	7358	48442	55800	0.868
August	157.1	148.7	50357	0	6804	48996	55800	0.878
September	153.7	152.2	51446	0	4994	49006	54000	0.908
October	134.9	139.9	47600	0	8402	47398	55800	0.849
November	112.4	120.0	41229	0	13562	40438	54000	0.749
December	117.8	128.5	44420	4	13394	42406	55800	0.760
Year	1853.9	1843.6	626300	12145	58726	598274	657000	0.911

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

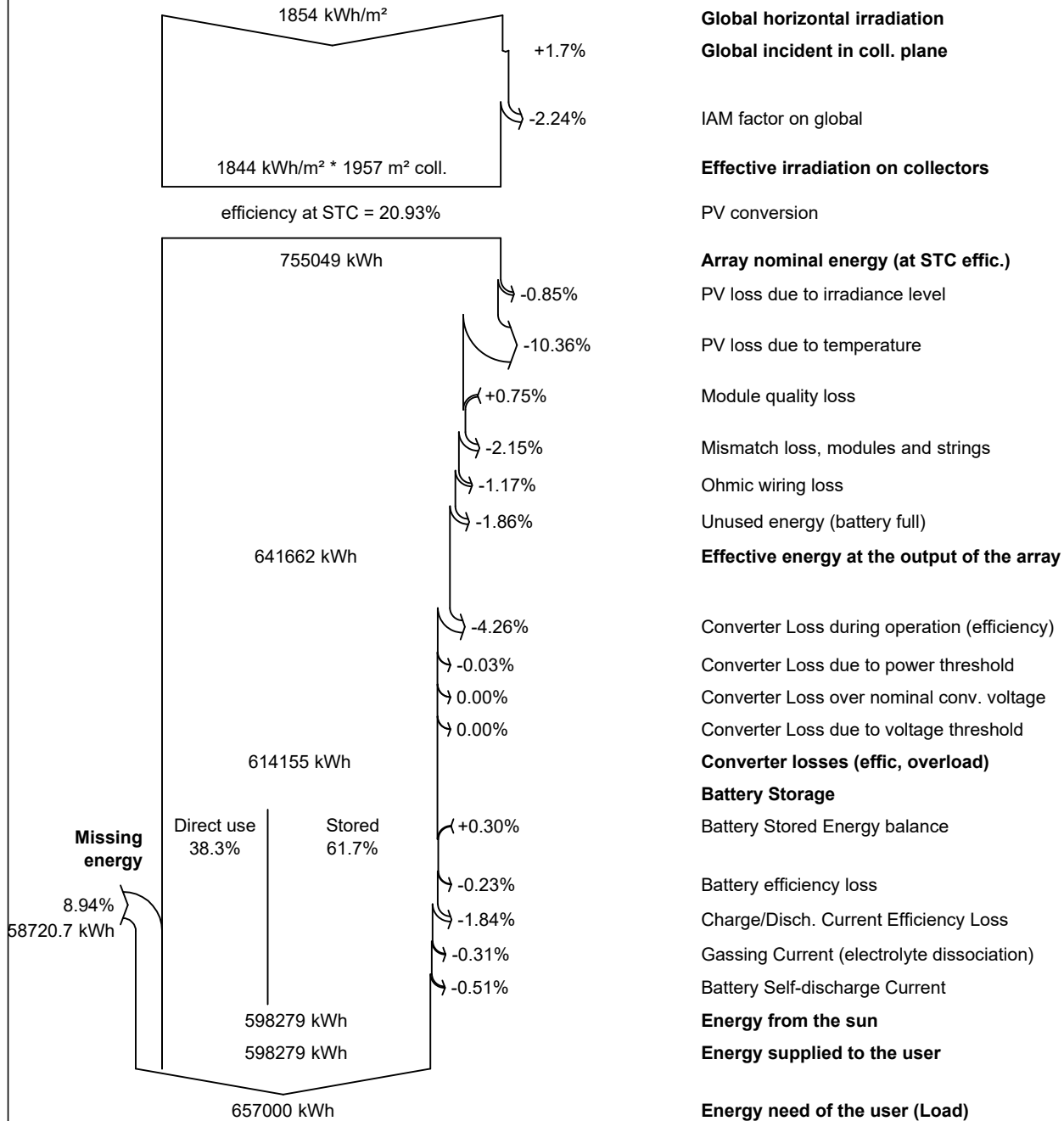
E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



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Loss diagram





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Predef. graphs

Daily Input/Output diagram

